# Sustainability Economics: Complete Lecture Notes

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# Course Overview & Syllabus Summary

What is the connection between economics and sustainability? When people talk about economics, often they are talking about something to do with money. How people acquire it and what they do with it. When people talk about sustainability, often they are talking about the environment, how it is changing, and whether or not those changes are good. Where do these two things come together? That intersection often seems to be about the way that people are acquiring and using money is changing the environment in ways that are not good.

# Part I

# Foundations & Basic Principles

# Chapter 1

# Introduction & Key Concepts

## 1.1 Overview & Objectives

The term *sustainability* implies continuity and balance. If something is sustainable, it can continue doing what it is doing. Unless it is doing nothing, what it is doing is interacting with other entities. This implies that, except for those things that are doing nothing, we should think about sustainability with a systems perspective.

A system is a collection of entities (or elements) and the connections between them. To understand a system, we need to understand the entities and their connections. Our goal in this course is to develop an understanding of the relationship between the economy and the environment by looking at the elements that make up the economy and their relationship with elements that make up the environment. To do so, we will proceed by building a model of a *complex adaptive social ecological system.*

The system we inhabit, and that we will try to understand, is complex. This means that the system as a whole exhibits properties that cannot be explained by looking at any element itself. It is adaptive in that it changes in response to internal and external pressures.

ey reading for this section: Chapter 1 of the required textbook.

**Systems**

- A system is a collection of elements and their connections.
- A social ecological system contains elements in both the social (human) and ecological (nature) domains, and has connections crossing between the domains.
- A system is complex when there are 'emergent' properties that cannot be predicted looking at elements in isolation.
- A system is adaptive if it continuously evolves or changes in response to internal and external forces.

**Systems**



### 1.1.1 Detailed Explanation of Concept 1

While the slide highlights the main points, it is important to note that the theoretical model relies on two primary assumptions...

**Mathematical Representation**

$$f(x) = \alpha + \beta x + \varepsilon$$

# Chapter 2

# Systems

## 2.1 Systems

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**Systems**



When we think about systems, we need to consider scale. If we consider the scale of ...

**Scales**

- What constitutes elements of a system depends on the physical scale at which we are contemplating the system.
  - At the atomic scale, atoms are the elements.
  - At the cosmic scale, galaxies are the elements.
  - What are the elements at the human scale?
- What the connections between elements are depends on the physical scale and the temporal scale chosen.
  - How long does it take for the connection to change something for the elements that we are interested in?
  - For the movement of atoms ...
  - For the movement of galaxies ...
  - For the human issues we are concerned about?

In both the natural and social systems, researchers try to identify patterns in the systems that they are studying. Those patterns emerge from the interactions between the elements that make up the system. When systems are large and complex, the discipline typically finds ways to describe the patterns without examining the detail of the interacting entities that leads to the pattern.

Cosmologists studying the movement of galaxies do not spend much time considering the movement of individual hydrogen atoms. However, it is the movement of those individual hydrogen atoms, under the influence of gravity, that leads to the formation of stars, and that leads to stars spinning together in galaxies.

Often, the disciplines have observed or thought deeply about the entities that they are interested in, and arrived at relationships - often mathematical - that represent the patterns that they observe.

In chemistry, the relationship $PV = nRT$ is known at the Ideal Gas Law. It is the relationship between pressure ($P$), volume ($V$) and temperature ($T$). The amount of gas in the volume is ($n$) in units of moles, and $R$ is the ideal gas constant, expressed the the appropriate units. A mole is Avagadro's number of atoms. This relationship expresses how temperature, which is a measure of how much energy atoms have, and the pressure that they are exerting within a particular volume. The physics of the situation is that the atoms are racing around, their speed reflecting the temperature, and smashing into the sides of the container that confines them. The more energy they have, the faster they are moving, and the harder they collide with the chamber. However, the equation is not representing the movement of those individual atoms. Rather, it is expressing this effect in a way that is relevant for things like the operation of a steam engine.

In the last few decades, computers have become powerful enough for us to represent the details of relationships like this from the basic building blocks. Agent based models are a class of such tools. One such agent based modeling environment we will use from time to time is NetLogo (https://www.netlogo.org/).

**Agent Based Models**

- Agent Based Models (ABMs) that interact virtual agents in virtual environments, ot observe how the system evolves.
- Agents can range from individual atoms to galaxies.
- Agents have often simple rules of behavior that they are programmed to follow.
- An ABM allows experiments with these agents, observing how the system that contains them evolves as a result of their interactions with other agents and the environment.
- ABMs allow properties of the system to emerge, properties that are not inherent to the agents that make up the system.
- ***emergence*** is an important characteristic of complex systems.

# Part II

# Foundational Systems

# Chapter 3

# Social Ecological Systems #1

When people who are concerned about the environment think of economics, they often conclude that it ignores the environment. The way introductory economics is taught often reinforces this, as if the environment is mentioned at all, it is a small topic, externalities.

There is a practical reason for this. During a presidential debate, former US president Bill Clinton said "It's the economy, stupid." [CONFIRM QUOTE]. He was emphasizing that what people are most worried about is their own situation. In a modern economy, your situation depends on your income and what it costs you to live. The environment is important, but is secondary to being able to put food on the table tonight, and having a roof under which that table is placed.

You are in this course because you are concerned about the environment, about sustainability. Therefore, lets start by building our understanding of sustainability economics from that environment.

**The Fish Pond**

- The Question: How does the population of fish in a pond change over time?
  - What time step?
  - One minute? - population changes small, unless major event.
  - One century? - population change in relation to long term events.
  - One year? - scale that matches seasons.
- The elements of a fish pond.
  - The fish (each fish, or fish as a population - biomass?).
  - The water (each molecule, or as a medium for the fish?).
  - The food (each food item - e.g phytoplankton, insects, other fish - or as a population - biomass?).
- The connections within the fish pond system.
  - Fish encountering food and eating.
  - Fish health and survival affected by state of the water.
  - Fish encounter each other and reproduce.

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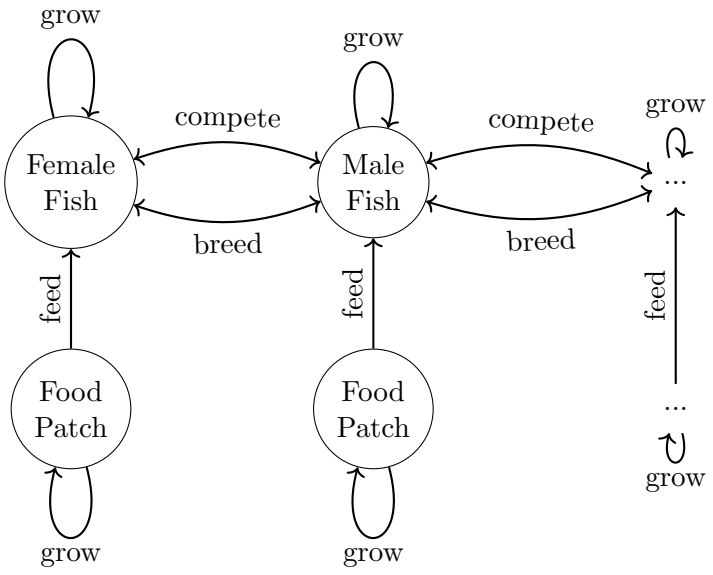
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Fish example: https://virtualbiologylab.org/ModelsHTML5/PopGenFishbowl/PopGenFishbowl.html

**The Fish Pond**



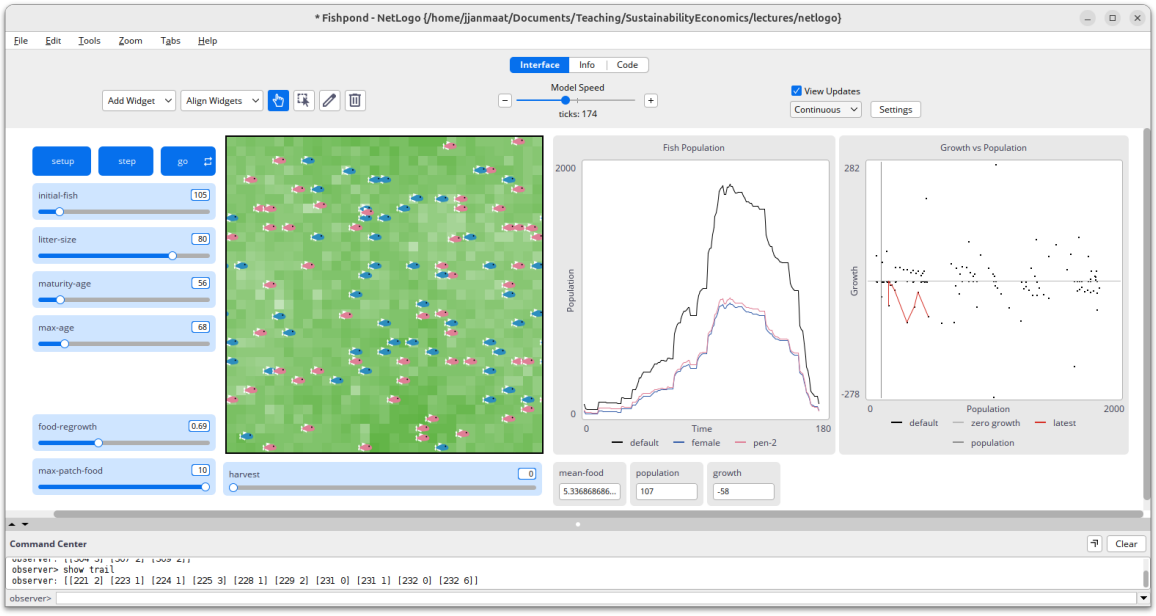
**The Fish Pond**

- What question(s) are we trying to answer with a model?
- What features of the elements (fish and food patches) and their interactions will lead to a sustainable system.
  - Sustainable = does not collapse to a system without fish.
- What level of harvest is sustainable?
  - Sustainable = harvest level that can continue without collapsing system to one without fish.
- How much detail do we need to model about the ecological system to identify the sustainable harvest level?
  - Can elements of fish pond system be treated together as one element of a social-ecological system?
  - Is the internal workings of the fish pond subsystem necessary to identify sustainable harvest levels?

**The Fish Pond**

- A NetLogo model.
- Fish (agents) swim around in pond with patches (environment) where they feed.
  - Fish: age to breeding, lifespan, how much food, impact of competition, breeding behavior, movement, ...
  - Food patch: how much food, growth rate of food.
- Lots of details that can be considered.
- Do we loose sight of the forest by focusing too much on the trees?

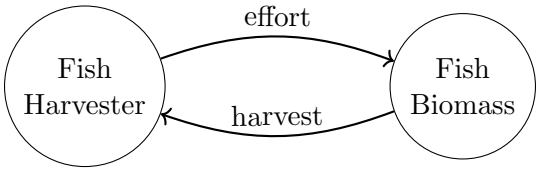
## The Fish Pond




## Fishing in the Pond

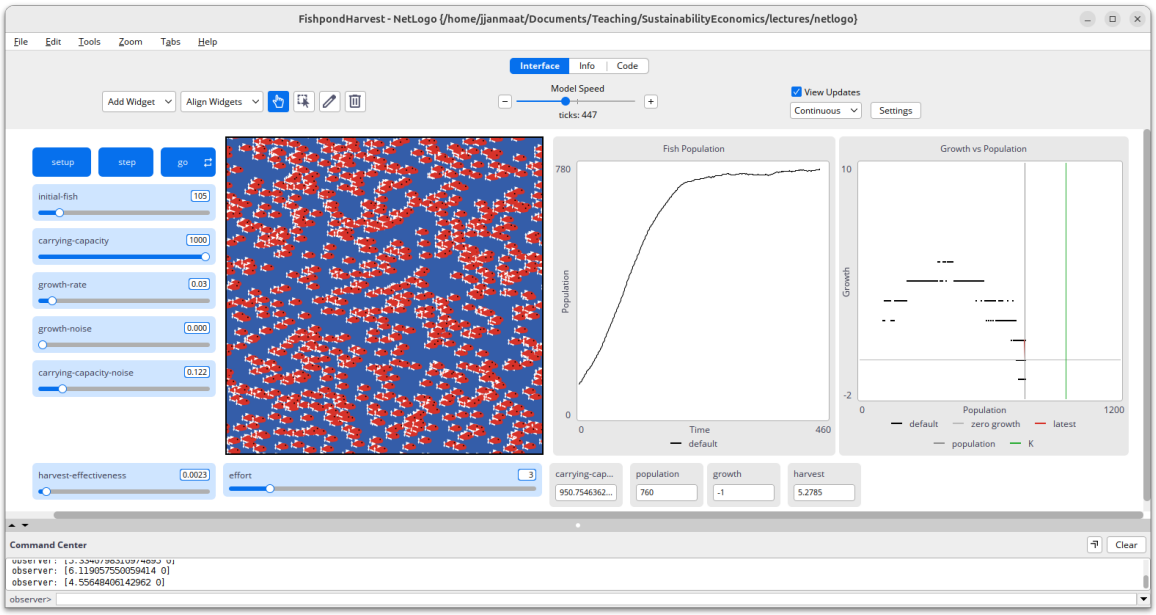
- Fish harvester chooses how much effort (e.g. time) to spend fishing.
- Most processes in pond not affected by amount of fishing effort.
- Effects of harvesting:
  - Reduces population of fish.
  - Reduces fish on fish competition - increasing growth.
  - Increases success of reproduction.
  - May force population so low that collapses.
- Harvest effort:
  - With same effort, more fish caught if more fish in pond.

## Fishing in the Pond




- What effort level will result in a sustainable harvest?

## Fishing in the Pond



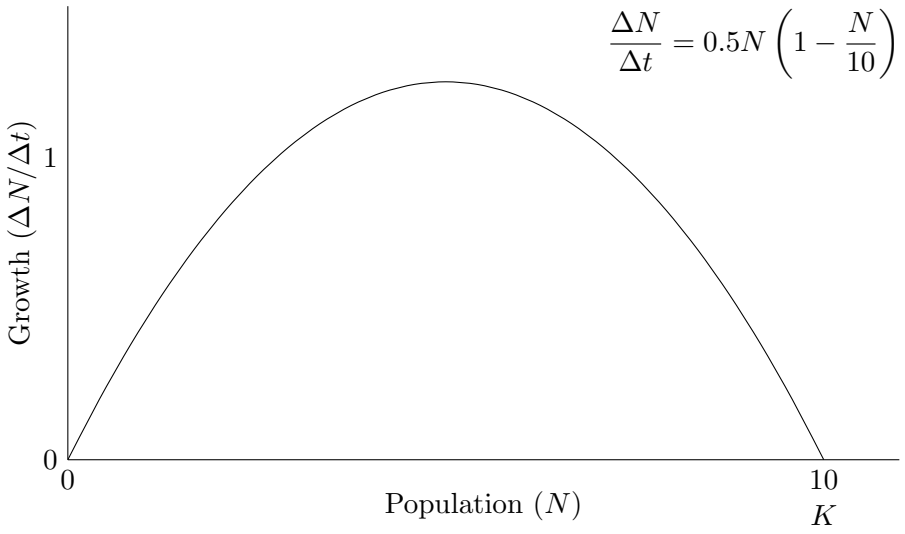

## Fishy Math

- Fish population modeled with logistic growth curve.

$$\frac{\Delta N}{\Delta t} = rN\left(1 - \frac{N}{K}\right) = \frac{\partial N}{\partial t} = \dot{N}$$

  - $\Delta N$ is change in fish population during time interval $\Delta t$.
  - $N$ is fish population.
  - $r$ is rate of growth of fish population.
  - $K$ is carrying capacity of the ecosystem represented by the model.

**Fishy Math**




**Fish Production Function**

- In economics, a ***technology*** refers to a process that transforms ***inputs*** into ***outputs***.
- A function that describes this process is called a ***production function***.
- In this example, effort is the input, and harvest is the output.

$$H = qEN$$

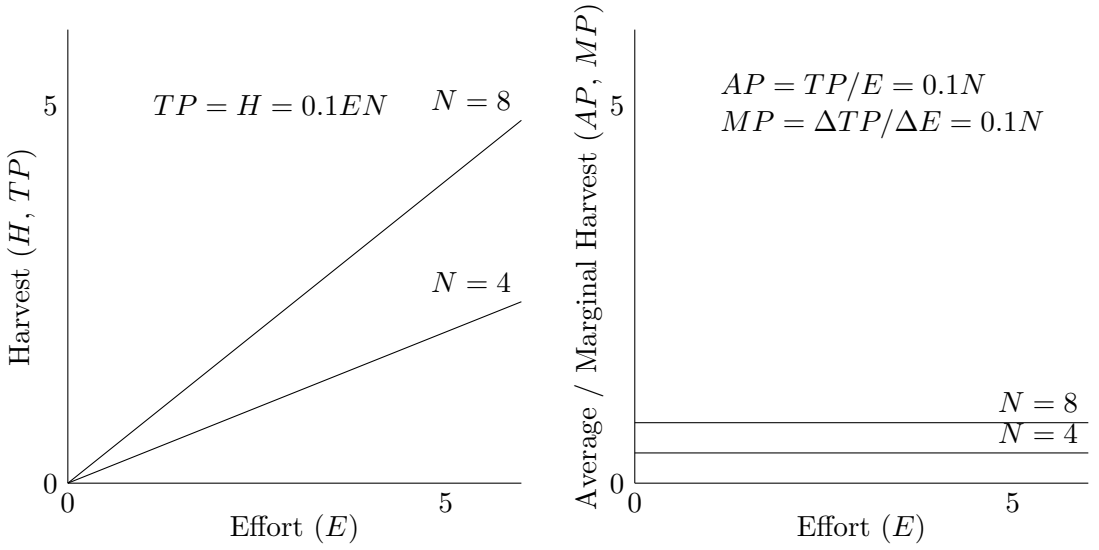
  - $H$ is the harvest.
  - $E$ is the effort.
  - $N$ is the population of fish.
  - $q$ measures how catchable the fish are.
- Note that the productivity of fishing effort depends on the size of fish population (the stock).
- A unit of effort $E$ harvests a fraction $q$ of the stock $N$.

**Fish Production Function**

- There are three measures that are commonly used in economics to describe aspects of the production process.

  1. ***Total Product (TP)*** is the total output at a particular input level $X$.
  2. ***Average Product (AP)*** is the total output at a particular level of input, divided by that level of input.
  $$AP = \frac{TP}{X}$$
  3. ***Marginal Product (MP)*** is, at a particular level of an input, the amount by which output will increase for a small increase in that input.
  $$MP = \frac{\Delta TP}{\Delta X} = \frac{\partial TP}{\partial X}$$
     - The marginal product is the slope of the total product at $X$.

**Fish Production Function**




The average and marginal product are the same for this production function. Given a particular fish population, each unit of effort catches the same number of fish. This would stop when there are no more fish left. For the range over which effort is shown in the figure, this does not happen.

Suppose the harvester is deciding how much effort to exert this period, which we will call a year. It is important to recognize that the choice of effort this year affects how many fish will be available to be caught next year. The fish that are left behind reproduce, and this reproduction adds to the supply of fish that can be caught next year.

We can throw some numbers at the example to make it a bit more concrete.

**Fish Production Function**

- Increasing effort increases harvest.
- Larger harvest means less fish left behind to reproduce.
- There is an ***opportunity cost*** to catching more fish today in terms of the lost contribution to growth of the fish stock in the future.

In the table below, the size of the fish stock, the amount harvested and the growth of the fish stock following harvest is shown for the first ten years, for effort levels $E = 2$ and $E = 3$.

The sequence within a year is that first fish reproduce and then the harvest is taken. What is left of the fish stock is the growth less the amount harvested.

The numbers in the table show a clear trend. For these two effort levels, the stock of fish and the harvest decrease over time. For the lower effort, the beginning harvest is lower. However, if this lower effort is maintained, then the lower harvest leads to a slower decline in the fish stock, and by the end of the ten years, the lower effort is resulting in a higher harvest than for the higher effort level.

If you look closely at the numbers, you can see that the differences from one year to the next are getting smaller. In each case, with the constant effort level, the system is converging on a ***steady state***. At the steady state, the size of the stock, the amount that is harvested, and the amount by which the fish stock grows are constant. In fact, the growth exactly matches the amount harvested. Each of these steady states is in effect sustainable. Which one do we like better?

**Fish Production Function**

$$H = 0.1EN \qquad \frac{\Delta N}{\Delta t} = 0.5N\left(1 - \frac{N}{10}\right) - H$$

| $t$ | $N$ | $E$ | $H$ | $\Delta N$ | $N$ | $E$ | $H$ | $\Delta N$ |
|---|---|---|---|---|---|---|---|---|
| 1 | 10.00 | 2 | 2.00 | 0.00 | 10.00 | 4 | 4.00 | 0.00 |
| 2 | 8.00 | 2 | 1.60 | 0.80 | 6.00 | 4 | 2.40 | 1.20 |
| 3 | 7.20 | 2 | 1.44 | 1.01 | 4.80 | 4 | 1.92 | 1.25 |
| 4 | 6.77 | 2 | 1.35 | 1.09 | 4.13 | 4 | 1.65 | 1.21 |
| 5 | 6.51 | 2 | 1.30 | 1.14 | 3.69 | 4 | 1.48 | 1.16 |
| 6 | 6.34 | 2 | 1.27 | 1.16 | 3.38 | 4 | 1.35 | 1.12 |
| 7 | 6.23 | 2 | 1.25 | 1.17 | 3.14 | 4 | 1.26 | 1.08 |
| 8 | 6.16 | 2 | 1.23 | 1.18 | 2.96 | 4 | 1.19 | 1.04 |
| 9 | 6.11 | 2 | 1.22 | 1.19 | 2.82 | 4 | 1.13 | 1.01 |
| 10 | 6.08 | 2 | 1.22 | 1.29 | 2.71 | 4 | 1.08 | 0.99 |

The following graphs show the progression of the stock as the continuing harvests change the size of the stock and the growth.

The system starts at the carrying capacity $K$. At the carrying capacity, the stock is steady. The new fish added to the population, the recruitment, is exactly matched by the deaths. There is no net growth of the stock.

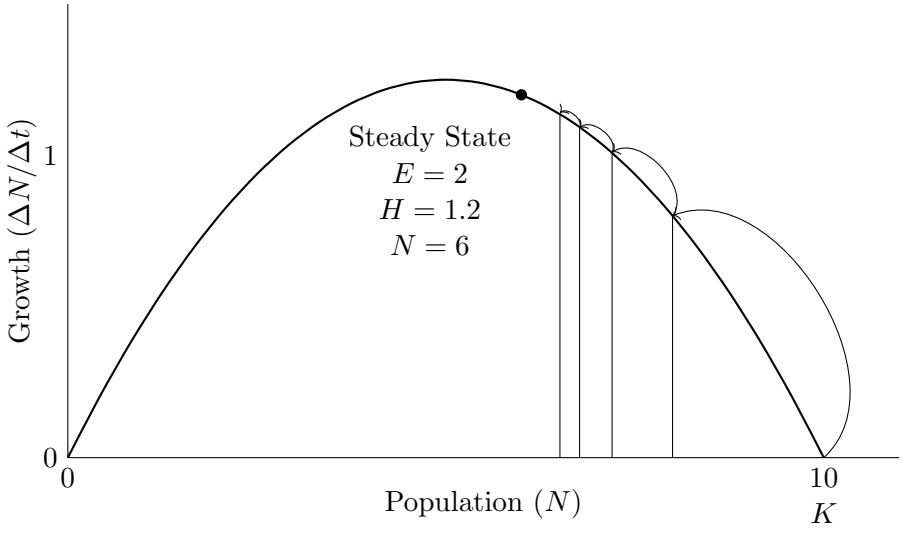
The first harvest reduces the stock by the amount of the harvest. When the effort is lower, this harvest is lower. As the stock is now below the carrying capacity, it has room to grow. That growth is represented by the height of the curve.

The second harvest captures a quantity of fish equal to all the growth and a bit more, reducing the stock further. This effect is larger for the case with the larger effort. With the stock reduced further, there is more capacity for growth.

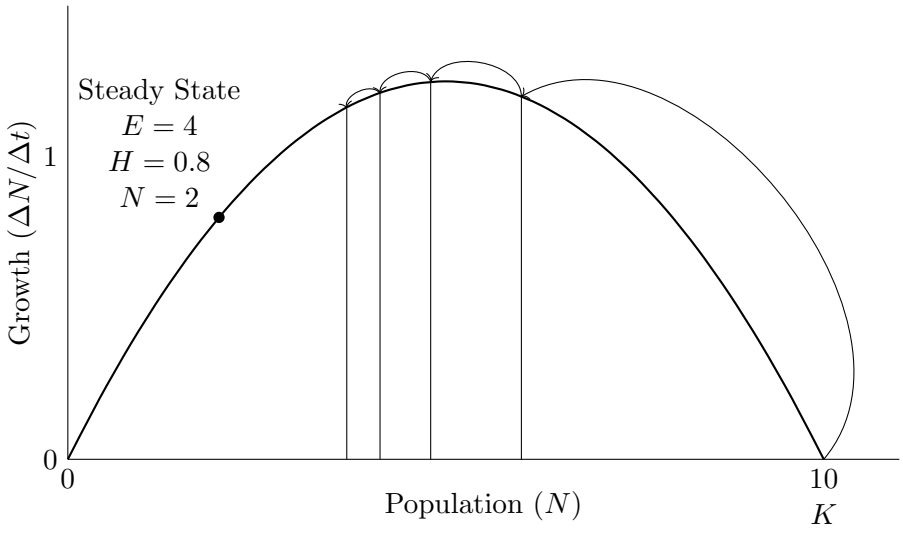
In the figures, things begin to diverge with the third harvest. When the effort is low, the change in the stock after the harvest leads to an increase in the growth of the stock. However, when the effort is high, the stock has been depleted so much that the further depletion results in a lower growth rate. We have moved the stock to a place where there is little competition for food, but it takes a lot of effort to find others to mate with.

If we continue with the same effort levels, the systems will converge on a steady state. The steady state is the level of the stock such that, given the effort level, that stock does not change any further. In these examples, the higher effort level leads to a lower steady state stock and a lower steady state harvest than for the lower effort level.

**Fish Production Function**



**Fish Production Function**




**Fishing Sustainably**

- Starting from fish population $N = 10$:
- Effort $E = 4$ leads to:
  - An immediate harvest of $H = 4$
  - A steady state with $N = 2$ and $H = 0.8$
- Effort $E = 2$ leads to:
  - An immediate harvest of $H = 2$
  - A steady state with $N = 6$ and $H = 1.2$
- Steady states are "sustainable", in that the activities can continue indefinitely.
- Is one steady state better?
- Can it be better to take big harvests in the early years, ending up with a lower sustainable harvest?

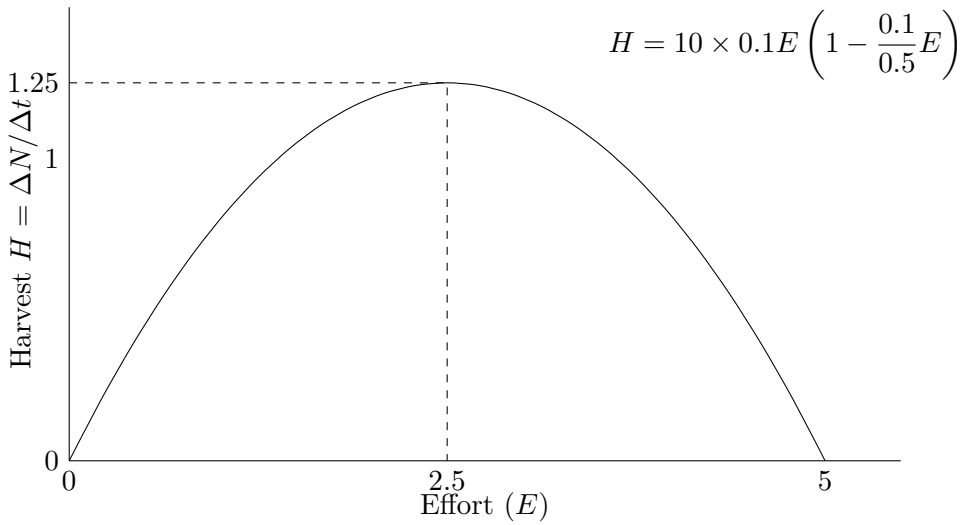
Each effort level corresponds to a unique steady state, or sustainable, stock and harvest level. These combinations are shown in the figure. Notice that the curve has the same shape as the stock and growth curve. However, the horizontal axis represents the effort devoted to fishing, and the vertical axis the harvest. The higher the effort level, the lower the corresponding steady state stock. The stock will be at its carrying capacity when there is no effort, when no fish are being harvested. With high enough effort, all the fish will be caught and the stock will be driven to zero.

Is an effort level that drives the stock to zero sustainable? Seems strange to consider. However, if the stock is zero and therefore the harvest is zero, this will continue to be the case into the future.

In a strange way, a fish stock that has been driven to extinction is now sustainable, as it is not changing any more.

This highlights that when we talk about sustainability, we are not just talking about a technical state of a system. Sustainability as a concept, in most of the ways that we use the idea, is not the same thing as a steady state. There is something more to it.

**Fishing Sustainably**




**Fishing Sustainably**

- Each effort level corresponds with a steady state "sustainable" stock size and harvest level.
- There is one effort level that corresponds to the maximum sustainable harvest.
  - In the fisheries literature, this is the ***maximum sustainable yield (MSY)***.
- When is it best to manage a resource to achieve the maximum sustainable yield?
- What is the ***function*** of this harvester and fish pond system?
- What would it mean for this harvester and fish pond system to be ***resilient***?

# Chapter 4

# Social Ecological Systems #2

Sustainability challenges exist because of the interaction between humans and the environment they inhabit.

Lets consider a system that incorporates humans and the environment.

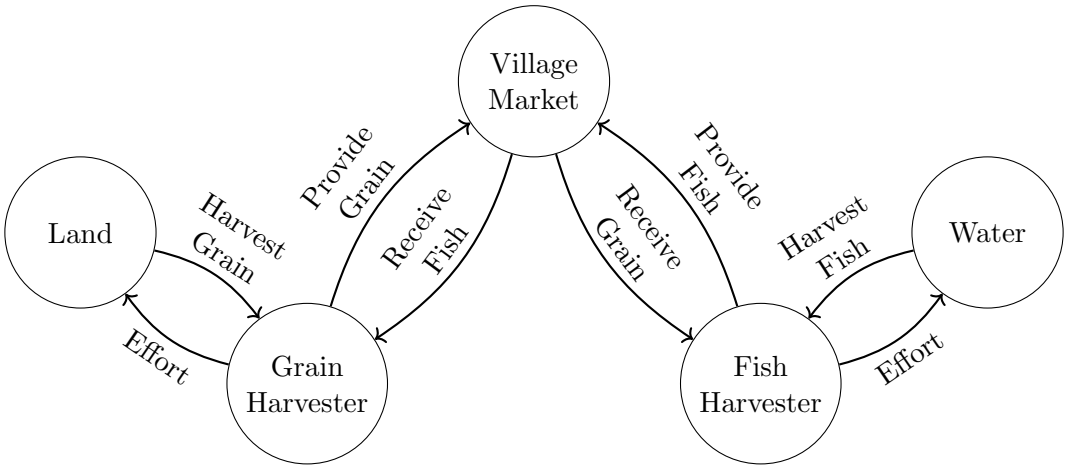
A simple example is a 'stylized' hunter gatherer model. In this case, our hunters will harvest fish, building on the model considered earlier.

**The Hunter Gatherer Village**

- The Question: ?
  - 1

An apparently simple mode can become complex pretty quickly.

**The Hunter Gatherer Village**

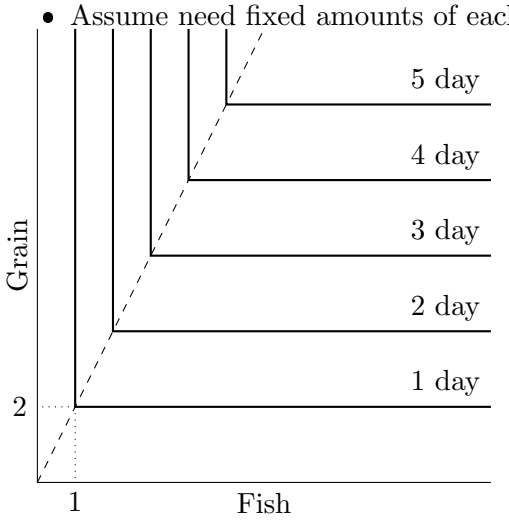


**The Hunter Gatherer Village**

- The grain subsystem
  - Grain grows on the land
  - Grain harvesters exert effort that translates into harvested grain.
- The fish subsystem
  - Fish grow in the water
  - Fish harvesters exert effort that translates into harvested fish.
- The Village system
  - Fish harvesters bring fish to the village, receive grain.
  - Grain harvesters bring grain to the village, receive fish.
  - How is the exchange of fish for grain managed?

**Human Ecology**

- Humans need food.
- Carbohydrates, fat and protein.
- Grain high in carbohydrates
- Fish provide fat and protein.
- Assume need fixed amounts of each.




Humans need food to survive. The basic components of that food are carbohydrates, fat and protein. There are various specific nutrients and micronutrients we need. So long as the fish and grain in our example provide sufficient quantities of these, we can focus on the balance of the three

main nutrients. For our model, we assume that the grain provides the carbohydrate requirements and the fish provide the fat and protein.

One way to think about this is in terms of the number of days that a person has enough grain and fish to survive. We can show this graphically using a contour map. The dimensions of the map are the amount of fish and the amount of grain that the person has in their possesses. Each contour line represents a day of survival.

The contours are 'L' shaped because survival for a day depends on consuming a minimum amount of each food source. Consider the contour representing one day of survival. The point where $q_F = 1$ and $q_G = 2$ is, in this example, the minimum amounts of fish and grain that are needed to survive one day. If the person has more fish, but only the two units of grain, they will not survive more than one day. Thus, with two units of grain, the contour for all larger quantities of fish always represents the same one day of survival.

The same logic applies when the person has one unit of fish. For any amount of grain that is more than two, the person cannot survive any additional days. Therefore the contour is vertical at the minimum amount of fish needed to survive one day.

For the mathematically inclined, the function for the contours is

$$d(q_F, q_G) = \min\left(\frac{q_F}{\alpha_F}, \frac{q_G}{\alpha_G}\right) \tag{4.1}$$

In this equation, $\alpha_F$ and $\alpha_G$ are, respectively, the amounts of fish and grain that the person needs to consume per day to survive that day. One can imagine that these amounts can differ by person. One can also imagine that if there is a change in technology that translates fish and grain into nutrients could also change this relationship. For example, a food preparation method that wastes less would reduce the amount needed per day of that food item.

What have we not considered? What would change if fish and grain decay? The fact that the kinks in the figure follow a straight line, and the spacing of the contours is always the same assumes that there is no decay. We know that grain can be stored far longer than raw fish. If the fish is not processed into something that can be stored - dried for example - then how would the figure change? What would happen if a share of the fish is lost each day? What would happen if no fish can be consumed after it is five days old?

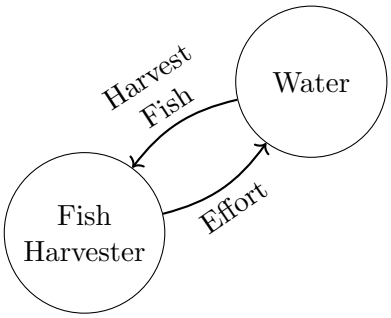
Another thing to consider is that while grain has far more carbohydrates than it does have fat and protein, it does have some. How would this change the graph? In those cases, it will not have the tight kink. Instead, it will have a bit of a curve, since there is some ability to substitute more grain for less fish.

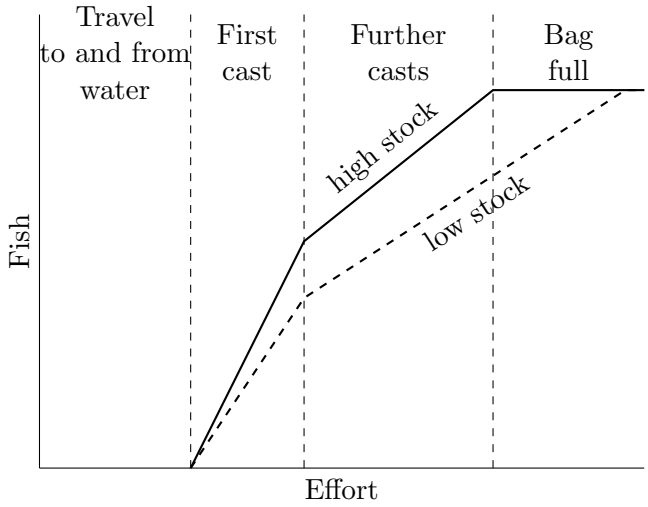
**The Fish Subsystem**

- Time path:
  1. Travel from village to water
  2. Cast net
  3. Collect fish from net until can't carry any more
  4. Travel back to village
  5. Exchange fish for grain
- Each step of time path requires effort.

If we consider the fish subsystem in relation to the village, we need to add a couple of elements that are not there when we just consider the relationship between effort and fish caught. The fish harvester also needs to spend time traveling between the village and the water where fishing is done.

**The Fish Subsystem**

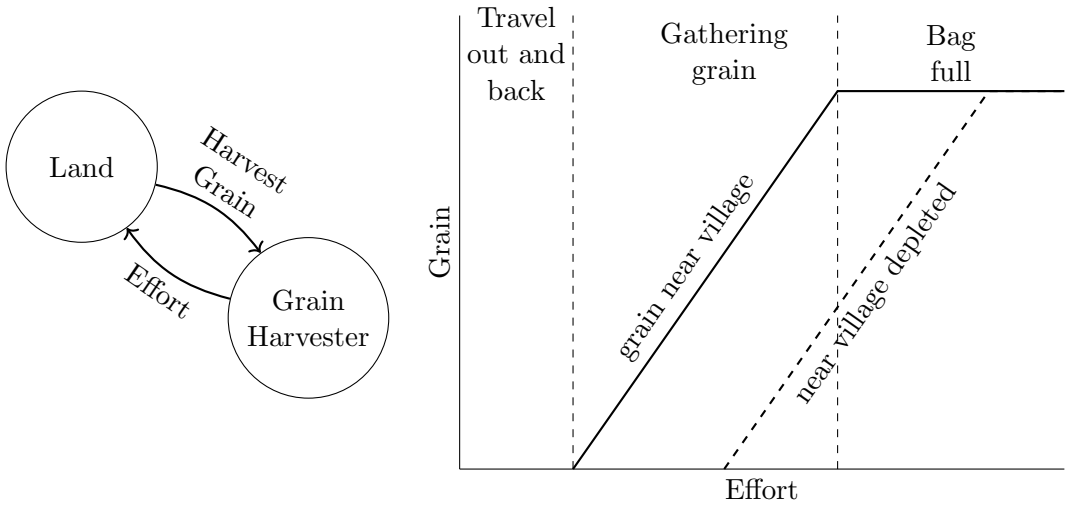



**The Grain Subsystem**

- Time path:
  1. Travel from village to closest patch of land with grain.
  2. Harvest grain from patch.
  3. Move to next patch and repeat until grain bucket full.
  4. Travel back to village
  5. Exchange fish for grain
- Each step of time path requires effort.

**The Grain Subsystem**

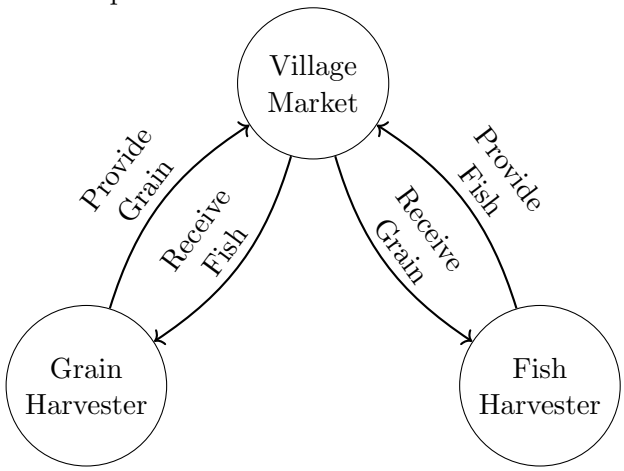



For the fish harvester, the travel distance before fish can be harvested is the same for every trip. For the grain harvester, the village is located among land that can produce grain. The harvester does not need to travel far if the land near the village has grain to harvest. However, if the land near the village has been depleted, then the grain harvester has to travel out from the village before any grain can be harvested, and has to travel further on the return trip.

How would things change if the amount of effort needed to travel to land that has grain? How would things change if the effort required to harvest a unit of grain is reduced? These are the impacts of technological change.

**The Village Market**

- How do fish and grain harvesters exchange their fish and grain?
- A collective? From each according to their means, to each according to their needs?
- A place where individuals trade for their own gain?



Figuring out how to represent the exchange of fish for grain is a tricky problem. What are the motivations that drive the decisions of these two people? Before delving into this problem, we can consider how their efforts may be combined to yield different combinations of fish and grain. For this exercise, imagine that there is enough fish and grain to share between the villagers so that they can live at least long enough to survive the trip to and from their respective harvest areas. If this is all they have, then they can travel out and back, without doing any harvesting or gathering, and upon return they would die, marked at $E_1$. Not ideal.

If there is sufficient food to allow them to survive beyond the travel time, then we can examine the combinations of fish and grain that can be produced by dividing the effort in different ways between fishing and gathering. In the figure, each line represents the combinations that can be produced for a given effort level. These curves are known as the ***production possibilities frontier (PPF)***. It is a frontier because it is what could be produced if all effort is used.

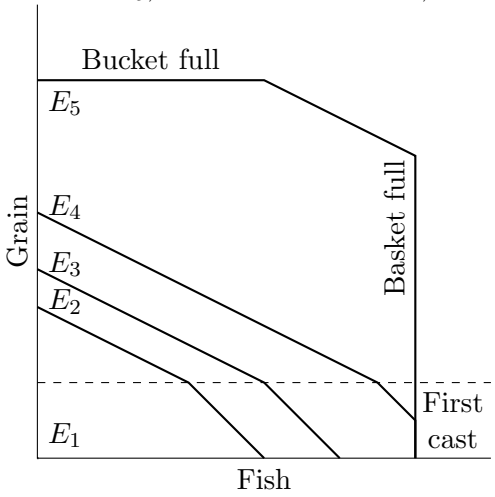
For the curves represented by $E_2$ and $E_3$, if the effort is reallocated from harvesting fish to gathering grain, the amount of fish harvested decreases and the amount of grain harvested increases.

With the PPF labeled $E_4$, we hit the limit of what the fish basket can hold. In effect, we 'throw away' part of the total effort, as we are dedicating it to catching fish, which we cannot take back to the village. Since our first cast is the most productive, we are in fact throwing away our most productive effort when running up against this constraint. If this seems strange, think of the order of events. The fish harvester cannot devote effort to later catches until they have made our first cast.

The PPF labeled $E_5$ corresponds to an effort level where one of the grain bucket or fish basket is full for most of the effort divisions. There is in effect a lot of wasted effort here, as it is being applied to activities which cannot result in any more grain or fish being brought back to the village. We might think of this wasted effort as leisure. If there is enough food in the village to sustain these harvesters and gatherers for this many days, maybe it would be better for them to stay in the village and enjoy themselves while eating the stockpiled food, heading out when they have enough food left to fill their basket or bucket and bring it back to the village.

**Production Possibilities**

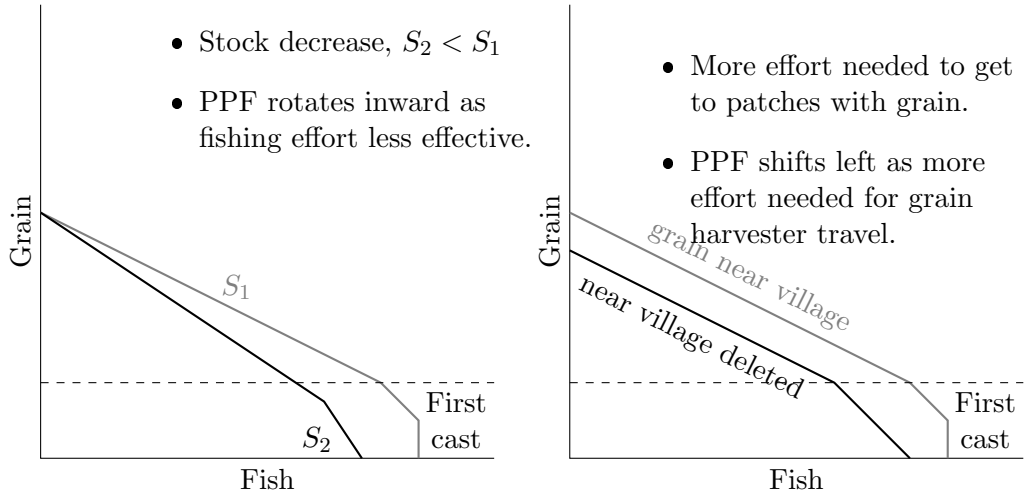
- Split total effort between fish and grain.
- $E_5 > E_4 > E_3 > E_2 > E_1$
- At $E_1$, no fish caught or grain gathered.
- At $E_5$, most effort not used, as bucket or basket full.




When the production functions were described above, some aspects relevant to sustainability were illustrate that do not show up here. For fishing, if harvesting reduces the fish stock, then the production function slope becomes flatter. It will take more effort, more days spent fishing, to fill up the fish basket. This shows up in the PPF as a rotation inward around the intercept with the Grain axis. The decline in the fish stock does not affect the amount of grain that can be gathered, if all the effort is devoted to gathering grain. However, the amount of grain production that is given up to harvest another fish increases.

When the grain available in patches near the village is depleted, more effort must be spent on travel. This reduces the total effort available for both activities, if we start from the same premise where we only examine how to reallocate effort after enough food is devoted to enabling both the fish harvester and the grain gatherer to travel to and from their respective activity locations. The PPF shifts left, lowering the intersection with the Grain axis. Since the total effort to divide is reduced, there is no longer enough effort to fill the fish basket, even when all the effort is put toward fishing.

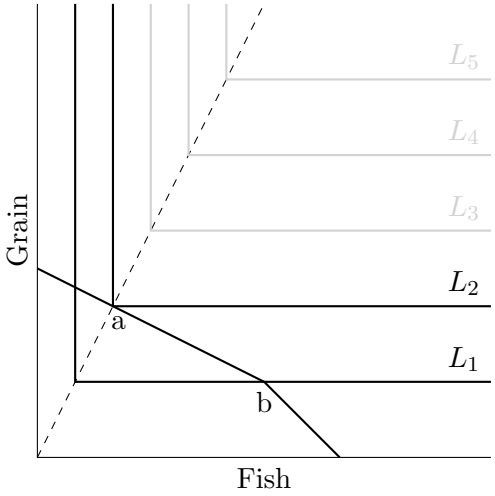
**Production Possibilities**




The production possibilities frontier tells us what combinations of fish and grain can be produced by the villagers, for different levels of effort. The biological reality of our villagers requires that the consume a particular bundle of fish and grain each day. If we put the biology together with the production possibilities frontier, we can see how to allocate the effort to provide the most days of life for the villagers.

**Production Possibilities**

- Lifespans $L_1$ and $L_2$ are attainable.
  - At **b** attain $L_1$.
  - At **a** attain $L_2$.
  - Same effort level.
- Lifespans $L_3$, $L_4$ and $L_5$ are not attainable.



Putting together the human needs and the production possibilities lets us identify something we might see as optimal. It seems reasonable to think that dividing up the resources that the villagers have in such a way as to maximize the number of days that they continue living would be a reasonable objective.